# BIOCHEMISTRY CUMULATIVE EXAM

## December 4, 1993

Choose a pentapeptide composed of 5 different amino acids.

(20 pts) 1. Draw the complete chemical formula for your peptide in its most favorable ionization state at pH 7.

(10 pts) 2. Estimate the net charge on your peptide at pH 11.

(10 pts) 3. How many different pentapeptides will have the same amino acid composition as yours? Justify your answer.

(10 pts) 4. Assume that your peptide is part of an alpha-helix. Show which groups in the peptide would be hydrogen bonded to each other in such a conformation.

(20 pts) 5. Describe an experimental method for distinguishing an alpha-helix from a beta-sheet. Do not choose a method that takes several months or years to complete. State the basis on which the method makes the distinction.

The following concerns a general protein of 100-200 amino acids.

(30 pts) 6. You wish to test whether the protein can be reversibly unfolded and refolded. How would you do such an experiment? Answer in as much detail as time and knowledge permit.